

Performance testing

Date	2 July 2026
Team ID	xxxxxx
Project Name	Opti crop: Smart Agricultural Optimization Engine
Maximum marks	5 marks

Performance testing

Step 1 : Testing overview

Field	Details
Testing Tool Used	postman
Type of testing	Load testing
Target Module /API	Crop recommendation prediction module (/predict)
Test Environment	Local system(flask)
Test date	14\07\26

Step 2: Test scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits crop recommendation request.	1	30	Crop recommendation displayed successfully.
2	Multiple users submit crop recommendation requests simultaneously.	10	60	Crop recommendations displayed successfully for all users.
3	Continuous crop prediction requests are processed without interruption.	20	120	Stable application performance without crashes or significant delay.
4	Invalid input values submitted (missing or incorrect values).	5	30	Appropriate validation/error message displayed to the user.

Step 3: performance test results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 seconds	Pass	Crop recommendation generated quickly.
2	Response Time (Max)	< 5 seconds	3.2 seconds	Pass	Response time remained within acceptable limit.

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
3	Throughput (Req/sec)	> 10 Req/Sec	17 Req/Sec	Pass	Application handled multiple prediction requests efficiently.
4	Error Rate	< 1%	0%	Pass	No errors observed during testing.
5	CPU Utilization	< 80%	41%	Pass	CPU usage remained within acceptable limits.
6	Memory Utilization	< 80%	46%	Pass	Memory usage remained stable throughout testing.

Step 4 : observation and analysis

The Smart Agricultural Production Optimization Engine (OptiCrop) performed efficiently during performance testing. The average response time remained below the target value, and the application successfully generated crop recommendations based on user-provided soil and environmental parameters. CPU and memory utilization remained within acceptable limits throughout testing, indicating efficient resource usage. No significant errors or application crashes were observed during continuous or concurrent requests. Overall, the system demonstrated stable performance and is suitable for providing accurate crop recommendations in a local Flask deployment environment.

Key Findings

- Application responded quickly to crop recommendation requests.
- Crop recommendations were generated accurately based on input parameters.
- The system remained stable during continuous and multiple user requests.
- Response time stayed within the expected performance limits.
- CPU and memory usage remained within acceptable thresholds.

Bottlenecks Identified

- Slight delay observed during the first prediction due to model initialization.
- Performance depends on the hardware configuration of the local system.
- High concurrent requests may slightly increase response time in the Flask development server.

Optimization Steps Taken

- Optimized the machine learning prediction workflow.
- Reduced unnecessary data processing before prediction.
- Implemented input validation to prevent invalid requests.

Step 5: Screenshots/Evidence

